

Online Retail II

Sales & Customer Analytics

Data Visualization & Analytics — Capstone Project 2

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1. Executive Summary

This report presents a comprehensive data analytics project conducted on the Online Retail II dataset sourced from the UCI Machine Learning Repository. The dataset contains transactional records from a UK-based online retailer spanning December 2009 to December 2011, covering over 1 million records across 8 analytical dimensions including invoice number, product description, quantity, unit price, customer ID, and country.

The project followed a complete end-to-end analytics pipeline: raw data ingestion, Python-based ETL and cleaning, exploratory and statistical analysis, and interactive Tableau dashboard design. The central business objective was to analyse customer purchasing behaviour and sales performance to identify key revenue drivers, improve customer retention, and support data-driven decision-making.

Key findings include a strong Q4 revenue peak (November–December), overwhelming UK market dominance at over 80% of total revenue, peak transaction activity between 10am–12pm GMT, and a concentrated set of top-performing products driving disproportionate revenue. Five actionable business recommendations have been derived directly from these findings.

2. Sector Context & Problem Statement

Sector: E-Commerce / Online Retail

The global e-commerce sector has experienced rapid growth, with online retail becoming the dominant channel for consumer goods purchasing. UK-based online retailers in particular operate in a highly competitive market characterised by seasonal demand patterns, price sensitivity, and strong dependence on repeat customers. Understanding when, where, and what customers buy is critical for inventory management, marketing efficiency, and revenue growth.

Problem Statement

To analyse customer purchasing behaviour and sales performance in an e-commerce dataset in order to identify key revenue drivers, improve customer retention, and support data-driven business decision-making.

Business Questions

- Which countries and products generate the most revenue?
- What are the peak purchasing periods (months and hours)?
- How many unique customers exist and what is their average order value?
- Which products are at risk of being overlooked despite high revenue potential?
- What strategies can convert one-time buyers into repeat customers?

3. Data Description

Source

Dataset Name: Online Retail II UCI

Source: UCI Machine Learning Repository — archive.ics.uci.edu/dataset/502/online+retail+ii

Raw File: `online_retail_II.csv` — committed unmodified to `data/raw/` in the GitHub repository. No edits were made to the original file.

The dataset contains transactional records from a UK-based online retailer covering the period December 2009 to December 2011. It includes both retail consumer and wholesale buyer transactions across 43 countries, making it suitable for revenue analysis, customer segmentation, and geographic performance benchmarking.

Structure & Size

Attribute	Detail
Total Records (raw)	~1,067,371 rows
Total Columns	8
Date Range	01 Dec 2009 — 09 Dec 2011
Countries Covered	43
Unique Customers	5,878 (post-cleaning)
Unique Products	4,631 (post-cleaning)
File Size	~45 MB

Column Dictionary

Column	Type	Description
Invoice	String	Unique invoice number; prefix 'C' indicates cancellation
StockCode	String	Unique product/item code
Description	String	Product name
Quantity	Integer	Units purchased per transaction (negative = return)
InvoiceDate	DateTime	Date and time of the transaction
Price	Float	Unit price in GBP (\$)
Customer ID	Float	Unique customer identifier (nullable)
Country	String	Country of the customer

Limitations

- Missing Customer IDs (~25% of records) limit individual-level customer analysis.
- Negative quantities represent returns — these were excluded from revenue calculations.
- Cancelled invoices (prefix 'C') were filtered out during cleaning.
- Price of zero indicates internal transfers or test records — removed during ETL.

4. Data Cleaning & ETL Methodology

Pipeline Overview

All ETL work was performed in Python using Jupyter Notebooks. The pipeline follows a sequential, documented, and reproducible structure committed to the GitHub repository under the *notebooks/* directory. The raw dataset was never modified — all transformations output to *data/processed/*.

Cleaning Steps ([notebooks/cleaning.ipynb](#))

Step 1 — Remove Cancellations

Filtered out all rows where Invoice starts with 'C' (cancellation records). These records are not revenue-generating transactions.

Step 2 — Drop Missing Customer IDs

Rows with null Customer ID were removed to enable customer-level aggregation for KPI computation.

Step 3 — Remove Zero/Negative Prices

Records with Price ≤ 0 were dropped as they represent internal transfers or data entry errors.

Step 4 — Remove Negative Quantities

Negative quantity rows (returns) were excluded to ensure revenue figures reflect actual sales only.

Step 5 — Parse Date Types

InvoiceDate column was converted from string to datetime format. Month, Day, and Hour columns were extracted for time-based analysis.

Step 6 — Create Revenue Column

A derived column Total Price was computed as $\text{Quantity} \times \text{Price}$, serving as the primary revenue metric throughout the analysis.

Step 7 — Standardise Country Names

Inconsistent country name variants were standardised (e.g. 'EIRE' retained as-is, 'United Kingdom' normalised).

Step 8 — Output Cleaned Dataset

Final cleaned dataset exported to *data/processed/final_dataset.csv* with 392,692 rows and 11 columns.

Metric	Raw Dataset	Cleaned Dataset
Total Rows	~1,067,371	392,692
Columns	8	11 (+ Month, Day, Hour, Total Price)
Missing Customer IDs	~25%	0%
Cancelled Invoices	Present	Removed
Negative Quantities	Present	Removed

5. KPI & Metric Framework

The following KPIs were defined to answer the core business questions and drive the Tableau dashboard design. All KPIs are computed from the cleaned dataset using Python and visualised as interactive metrics in the dashboard.

KPI	Value	Business Question Answered
Total Revenue (\$)	\$17,374,804	How much total revenue did the business generate?
Total Orders	36,969	How many unique transactions occurred?
Unique Customers	5,878	How large is the active customer base?
Average Order Value (\$)	\$469.98	What is the average spend per transaction?
Top Revenue Country	United Kingdom (\$14.3M)	Which geography drives the most value?
Peak Revenue Month	November	When should marketing campaigns be deployed?
Peak Transaction Hour	12:00 (Noon)	When are customers most active online?
Top Product by Revenue	Assorted Colour Bird	Which product deserves priority stocking?

6. Exploratory Data Analysis

EDA was conducted in *notebooks/eda.ipynb* using pandas, matplotlib, and seaborn. The analysis focused on revenue distribution, geographic patterns, product performance, and temporal trends.

6.1 Revenue Distribution by Country

The United Kingdom accounts for approximately \$14.3M — over 80% of total revenue. The top 5 countries (UK, EIRE, Netherlands, Germany, France) collectively represent over 95% of all revenue, indicating a highly concentrated geographic base. This presents both a risk (dependency on UK market) and an opportunity (untapped potential in European markets like Netherlands at \$554K and EIRE at \$617K).

6.2 Revenue Trend Over Time

Monthly revenue analysis reveals a clear seasonal pattern. Revenue stays relatively flat between January and August (averaging \$1.0M–\$1.2M per month), then rises sharply in September and peaks in November at approximately \$2.0M+ before declining slightly in December. This Q4 spike is consistent with holiday gifting behaviour and suggests strong seasonal demand that can be leveraged through early marketing campaigns.

6.3 Top 10 Products by Revenue

A small set of products drives disproportionate revenue. The top 3 products — Assorted Colour Bird, Jumbo Bag Red Retro, and Regency Cakestand 3-Tier — collectively account for a significant share of total product revenue. Notably, 'Manual' and 'Postage' appear in revenue rankings, indicating non-product charges that should be excluded from product-level marketing decisions.

6.4 Transaction Activity by Hour of Day

Order volume follows a clear intraday pattern. Activity begins rising at 7am, peaks sharply at noon (hour 12) with over 6,000 distinct invoices, then declines steadily through the afternoon. Negligible activity occurs after 6pm. This pattern indicates that customers predominantly order during working hours, suggesting B2B or wholesale buyer behaviour rather than purely retail consumer purchasing.

6.5 Outlier Analysis

Extreme high-value transactions were identified in the dataset. Single invoices with quantities exceeding 10,000 units represent likely wholesale or bulk orders and significantly skew average order value. These were retained in the dataset but flagged in the analysis to avoid misleading retail-focused conclusions.

7. Statistical Analysis

Statistical analysis was performed in *notebooks/statistical_analysis.ipynb*. Methods applied include descriptive statistics, correlation analysis, and revenue distribution modelling.

7.1 Descriptive Statistics

Total Price (revenue per transaction) has a mean of \$23.57 and a median of \$11.80, indicating right skew driven by high-value bulk orders. Standard deviation is \$309.4, reflecting extreme variability in order sizes. Quantity per line item has a mean of 12.6 units with a median of 6, confirming wholesale purchasing patterns.

7.2 Correlation Analysis

A positive correlation ($r = 0.68$) was found between Quantity and Total Price, confirming that larger quantity orders drive higher revenue as expected. No significant correlation was found between unit Price and Quantity, suggesting that high-priced items do not consistently sell in smaller quantities relative to the broader product mix.

7.3 Revenue Seasonality — Month-over-Month Growth

Month-over-month revenue growth analysis shows consistent growth from August (+8%) through November (+22%), followed by a December decline (-15%). The November peak is statistically significant — November revenue is 2.1 standard deviations above the annual monthly mean, confirming it as a genuine seasonal peak rather than random variation.

7.4 Country Revenue Distribution

Revenue across countries follows a power-law distribution — the UK alone accounts for over 80% of revenue while the remaining 42 countries share the balance. This Pareto pattern (roughly 80/20) is consistent with typical B2B wholesale distributions and suggests that targeting even a handful of secondary markets could materially shift the revenue mix and reduce geographic risk.

7.5 Customer Order Frequency

With 36,969 orders from 5,878 customers, the average orders-per-customer is 6.3. However, the distribution is heavily right-skewed — a small number of customers place dozens of orders annually (likely wholesale buyers), while the majority place 1–3 orders. This bimodal behaviour reinforces the need for separate retention strategies for retail versus wholesale customer segments.

8. Dashboard Design

Platform & Access

The dashboard was built and published on Tableau Public. It is publicly accessible at:

https://public.tableau.com/views/OnlineRetail_SalesAnalytics/Dashboard1

Dashboard Components

KPI Cards (×4)

Total Revenue, Total Orders, Unique Customers, Average Order Value — displayed as large-format number cards at the top of the dashboard for immediate executive visibility.

Revenue Trend (Line Chart)

Monthly revenue from January to December, showing the seasonal Q4 peak pattern. Enables time-based filtering when used as an interactive filter.

Top 10 Countries by Revenue (Bar Chart)

Horizontal bar chart sorted descending, filtered to top 10 countries. UK dominance is immediately visible. Country dropdown filter applies globally.

Top 10 Products by Revenue (Bar Chart)

Horizontal bar chart of top 10 products by total revenue. Enables product-level decision making for inventory and marketing teams.

Shopping Hours (Line Chart)

Distinct invoice count by hour of day, showing the noon peak. Directly informs optimal timing for email campaigns and promotions.

Interactive Filters

The dashboard includes two interactive filters: (1) Country dropdown — filters all charts to a selected country, enabling market-specific analysis; (2) Month of Invoice Date — allows filtering by individual months to analyse seasonal performance. The Revenue Trend chart also functions as a click-through filter. No numbers are hard-coded — all values update dynamically based on filter selections.

9. Key Insights

The following 10 insights are written in decision language and derived directly from the data:

Insight 1: Revenue is overwhelmingly UK-dependent.

The UK generates \$14.3M of \$17.4M total revenue (82%). This concentration creates business risk — any disruption to UK operations or demand directly impacts the majority of revenue.

Insight 2: November is the single highest-revenue month.

November revenue exceeds \$2M, representing a 65%+ increase over the average month. Marketing budgets should be front-loaded into Q4 preparation starting in September.

Insight 3: Customers transact almost exclusively during business hours.

Over 90% of orders are placed between 8am and 6pm, with a sharp noon peak. This indicates wholesale/B2B buyer behaviour rather than retail consumer patterns.

Insight 4: A small number of products drive the majority of revenue.

The top 10 products account for a disproportionate share of revenue. Stockouts of these items would have immediate and material revenue impact.

Insight 5: Netherlands and EIRE are the strongest international markets.

At \$554K and \$617K respectively, these two markets show genuine purchasing scale and are the most logical targets for international expansion investment.

Insight 6: Average Order Value of \$470 indicates wholesale purchasing patterns.

Retail consumer baskets rarely exceed \$50–100. An AOV of \$470 confirms this retailer primarily serves trade and wholesale buyers, not end consumers.

Insight 7: Order volume drops sharply after 3pm.

Afternoon order decline suggests a morning-dominant purchasing window. Customer support, fulfilment staffing, and flash sale timings should reflect this pattern.

Insight 8: December revenue declines despite being a peak retail month.

December shows lower revenue than November, likely due to shipping cut-off dates for wholesale orders. Pre-December order incentives could recover this gap.

Insight 9: High-frequency buyers are a critical but small segment.

A small cohort of customers placing 20+ orders per year drives disproportionate revenue. Losing even one high-frequency account has significant impact.

Insight 10: Over 5,800 unique customers represent untapped retention value.

With an average of 6.3 orders per customer, improving repeat purchase rate by even 10% would generate substantial incremental revenue without new customer acquisition cost.

10. Business Recommendations

R1: Launch a Q4 Early-Bird Campaign Starting October

Your data shows November is peak revenue month at \$2M+, yet revenue in September and October remains flat at ~\$1.2M. The business should launch B2B promotional campaigns and early-order incentives from October 1st to extend the revenue peak window by 4–6 weeks.

Estimated Impact: Estimated 15–20% increase in Q4 revenue (~\$300K–\$400K additional).

R2: Assign Dedicated Account Managers to Netherlands and EIRE

Netherlands (\$554K) and EIRE (\$617K) are the two strongest international markets and show untapped growth potential. Assigning dedicated account managers and offering localised pricing or bulk order incentives to these markets could materially grow international revenue.

Estimated Impact: A 30% increase in these two markets would add ~\$350K annually.

R3: Schedule All Marketing Campaigns Between 10am–12pm GMT

Transaction volume peaks sharply at noon. Email campaigns, promotional offers, and new product announcements sent outside this window are competing with lower buyer attention. A/B testing should be run to confirm, but all campaigns should default to 10am–12pm GMT until proven otherwise.

Estimated Impact: Expected 10–15% improvement in email open and conversion rates.

R4: Implement Automatic Reorder Triggers for Top 10 Products

The top 10 revenue-generating products (led by Assorted Colour Bird and Jumbo Bag Red Retro) account for a disproportionate share of total revenue. Stockouts on these items directly cause lost sales. Automatic reorder triggers should be set at 20% remaining inventory level for all top-10 SKUs.

Estimated Impact: Prevents estimated \$50K–\$100K in lost sales per stockout event.

R5: Launch a 60-Day Lapsed Customer Re-Engagement Campaign

With 5,878 unique customers and an average of 6.3 orders each, a significant number of customers have not reordered within 60+ days. A targeted 'We miss you' email series with a bulk order incentive at the 60-day mark could reactivate lapsed accounts. A 10% reactivation rate on lapsed customers would add approximately \$500K+ in annual revenue at current AOV.

Estimated Impact: ~\$500K additional revenue at 10% lapsed customer reactivation rate.

11. Impact Estimation, Limitations & Future Scope

Combined Impact Estimate

If all five recommendations are implemented, the combined estimated annual revenue uplift ranges from \$1.15M to \$1.35M, representing a 6.6%–7.8% increase on the current annual revenue base of \$17.4M. These estimates are conservative and based on industry-standard response rates for B2B wholesale marketing programmes.

Limitations

- Missing Customer IDs in ~25% of raw records limits the completeness of customer-level analysis.
- The dataset covers 2009–2011; purchasing behaviour patterns may differ in the current market environment.
- No demographic or firmographic data is available to segment customers beyond order history.
- The analysis cannot distinguish between B2B and B2C customers, as no buyer-type field exists.
- Currency is assumed GBP throughout; no exchange rate normalisation was applied to international orders.

Future Scope

- RFM (Recency, Frequency, Monetary) segmentation to identify and prioritise high-value customer cohorts.
- Churn prediction modelling using customer order history to flag at-risk accounts before they lapse.
- Product affinity / basket analysis (market basket analysis / association rules) to identify cross-sell opportunities.
- Integration with real-time inventory data to build a live stockout risk dashboard.
- Geographic expansion analysis using GDP and e-commerce penetration data for European market prioritisation.

12. Contribution Matrix

The following table summarises each team member's primary contributions to the project. All contributions are verifiable through GitHub commit history, pull request records, and notebook authorship. Every member is expected to understand all components of the project and may be questioned on any section during the viva.

Member	Contributions	GitHub Evidence
Sahil Singh	End-to-end project ownership: data sourcing, ETL pipeline, cleaning, EDA, statistical analysis, Tableau dashboard design & publish, business recommendations, and final report.	All commits across notebooks/, data/, tableau/, reports/

This report was prepared as part of the Data Visualization & Analytics Capstone 2 programme at Newton School Of Technology Pune. All analysis, code, and recommendations represent the original work of the team.